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Automate Cybersecurity Tasks with Python

Module 1
Introduction to Python

Module 2
Write effective Python code

Module 3
Work with strings and lists

Module 4
Python in practice

Python for automation

- ✓ Welcome to module 4
  Video • 1 min
- ✓ Automate cybersecurity tasks with Python
  Video • 3 min
- ✓ Automating Tasks in CI/CD
  Reading • 10 min
- ✓ Essential Python components for automation
  Reading • 8 min
- ✓ Clancy: Continual learning and Python
  Video • 2 min
- ✓ Test your knowledge: Python and automation
  Practice Assignment • Grade: 100%

Work with files in Python

- ✓ Access a text file in Python
  Video • 3 min
- ✓ Import files into Python
  Reading • 8 min
- ✓ Parse a text file in Python
  Video • 2 min
- ✓ Work with files in Python
  Reading • 8 min
- ✓ Activity: Import and parse a text file
  Lab • 40 min
- ✓ Exemplar: Import and parse a text file
  Lab • 20 min
- ✓ Develop a parsing algorithm in Python
  Video • 6 min
- ✓ Activity: Create another algorithm
  Lab • 50 min
- ✓ Exemplar: Create another algorithm
  Lab • 20 min
- ✓ Portfolio Activity: Update a file through a Python algorithm
  Graded Assignment • Grade: 100%
- ✓ Portfolio Activity Exemplar: Update a file through a Python algorithm
  Reading • 4 min
- ✓ Test your knowledge: Work with files in Python
  Practice Assignment • Grade: 100%

Debug Python code

- ✓ Debugging strategies
  Video • 6 min
- ✓ Matt: Learning from mistakes
  Video • 3 min
- ✓ Apply debugging strategies
  Video • 5 min
- ✓ Explore debugging techniques
  Reading • 8 min
- ✓ Activity: Debug Python code
  Lab • 35 min
- ✓ Exemplar: Debug Python code
  Lab • 20 min
- ✓ Test your knowledge: Debug Python code
  Practice Assignment • Grade: 100%

Review: Python in practice

- ✓ Coach dialogue: Automate security tasks and parse files with Python
  Dialogue • 15 min
- ✓ Wrap-up
  Video • 1 min
- ✓ Reference guide: Python concepts from module 4
  Reading • 4 min
- ✓ Glossary terms from module 4
  Reading • 4 min
- ✓ Module 4 challenge
  Graded Assignment • Grade: 100%

Congratulations on completing Course 7!

- ✓ Reflect and connect with peers
  Reading • 2 min
- ✓ Course wrap-up
  Video • 2 min
- ✓ Reference guide: Python concepts from Course 7
  Reading • 4 min
- ✓ Course 7 glossary
  Reading • 2 min
- ○ **Get started on the next course**
  Reading • 4 min

Get started on the next course

1:56 / 2:57 1x

Congratulations on completing Course 7 of the Google Cybersecurity Certificate: **Automate Cybersecurity Tasks with Python**! In this part of the program, you learned about writing effective Python code to automate cybersecurity tasks. You started with foundational components of the Python language, including data types, variables, conditional statements, and iterative statements. Then you built on those skills by learning how to incorporate functions into your code and work with string and list data. You also explored modules and libraries, as well as guidelines for improving your code's readability. Finally, you used your Python skills to import and parse files and to develop algorithms that automate security tasks.

The Google Cybersecurity Certificate has nine courses:

Foundations of Cybersecurity
Play It Safe: Manage Security Risks
Connect and Protect: Networks and Network Security
Tools of the Trade: Linux and SQL
Assets, Threats, and Vulnerabilities
Sound the Alarm: Detection and Response
Automate Cybersecurity Tasks with Python
Put It to Work: Prepare for Cybersecurity Jobs

1. **Foundations of Cybersecurity** — Explore the cybersecurity profession, including significant events that led to the development of the cybersecurity field and its continued importance to organizational operations. Learn about entry-level cybersecurity roles and responsibilities.
2. **Play It Safe: Manage Security Risks** — Identify how cybersecurity professionals use frameworks and controls to protect business operations, and explore common cybersecurity tools.
3. **Connect and Protect: Networks and Network Security** — Gain an understanding of network-level vulnerabilities and how to secure networks.
4. **Tools of the Trade: Linux and SQL** — Explore foundational computing skills, including communicating with the Linux operating system through the command line and querying databases with SQL.
5. **Assets, Threats, and Vulnerabilities** — Learn about the importance of security controls and developing a threat actor mindset to protect and defend an organization's assets from various threats, risks, and vulnerabilities.
6. **Sound the Alarm: Detection and Response** — Understand the incident response lifecycle and practice using tools to detect and respond to cybersecurity incidents.
7. **Automate Cybersecurity Tasks with Python** — Explore the Python programming language and write code to automate cybersecurity tasks. *(This is the course you just completed. Well done!)*
8. **Put It to Work: Prepare for Cybersecurity Jobs** — Learn about incident classification, escalation, and ways to communicate with stakeholders. This course closes out the program with tips on how to engage with the cybersecurity community and prepare for your job search.
9. **Accelerate Your Job Search with AI** — Gain practical job search strategies and learn how to leverage AI tools (like Gemini and NotebookLM) to uncover your most valuable skills, create a job search plan, manage your applications, and practice for interviews as you navigate your path to your next role.

Now that you have completed this course, you're ready to move on to the next course:
Put It to Work: Prepare for Cybersecurity Jobs.

Keep up the great work!

Mark as completed

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